

Scalability & Future Plan

Date	08.10.2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI responses depend on an external service; the workout-recommendation call took about 7.8 seconds.	Slow user experience and possible timeouts when many users request AI output together.	High
2	The API runs only locally (localhost:5000) with a local MongoDB instance; no hosted deployment or Dockerfile.	Not accessible to external users; limited availability and difficult to scale.	High
3	No caching, rate limiting or automated load handling on the current single server instance.	Risk of overload, high AI usage cost and abuse of endpoints.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js / Express instance on a local machine.	Deploy to a cloud platform, run multiple instances with PM2 / Docker containers behind a load balancer.
2	Data Storage	Local MongoDB (Mongoose models for users and workouts).	Move to managed MongoDB Atlas with replica sets, indexes on user and workout queries, and regular backups.
3	Performance	AI calls handled synchronously; responses take several seconds.	Cache repeated AI results, set request timeouts, use a background job queue for long AI tasks, and add response compression.
4	Security	JWT authentication, .env for secrets, .gitignore for the .env file.	Add rate limiting, Helmet security headers, stricter input validation, HTTPS, refresh tokens and a secrets manager.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment with Docker, hosted MongoDB Atlas, automated API tests in a CI pipeline.	1–2 months	Public access, stable environment and easier evaluation.
Phase 3	Caching, rate limiting and security hardening; richer AI insights using workout history trends.	3–4 months	Faster responses, safer API and more personalised insights.
Phase 4	Web / mobile front end, goal tracking, progress charts and wearable-device data integration.	5–6 months	Complete fitness-tracking product with a larger user base.